UNIT-I

1. Find the rank of Matrix $\begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}$

2. Find the rank by reducing to normal form $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$

3. Find the inverse of the matrix $\begin{bmatrix} 1 & -2 & -3 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ by using elementary row operations

4. Solve the following system of equations using Gauss-Seidal iteration method up to three decimal accuracy

$$10x + y + z = 12$$
$$2x + 10y + z = 13$$
$$2x + 2y + 10z = 14$$

5 Find all the solutions of the system of equations

$$x + 2y - z = 0$$
$$2x + y + z = 0$$
$$x - 4y + 5z = 0$$

6. Test for the consistency of the system of linear equations

$$x + y + 2z = 4$$
$$2x - y + 3z = 9$$
$$3x - y - z = 2$$

If consistent solve them

Unit 2

1. Verify Cayley–Hamilton theorem for the matrix

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$ and hence find and hence find A^{-1}

2 Verify Cayley- Hamilton theorem for the matrix A and hence find A^{-1} and A^4 given $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$

3 Determine the modal matrix which transforms the matrix $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$ Into diagonal form and hence calculate A^4

4. Reduce the Quadratic form $3x^2+5y^2+3z^2-2yz+2zx-2xy$ to the canonical form by orthogonal transformation and hence find rank, Index, signature and nature of the Quadratic form

5 Reduce the quadratic form $2x^2+2y^2+2z^2-2xy-2zx-2yz$ to sum of squares form by orthogonal transformation. Also find its rank, Index, signature and nature.

Unit3:

1. State Rolle's theorem and verify Rolle's theorem for the function

$$f(x) = (x+2)^3(x-3)^4 \quad x \in [-2, 3]$$

2. If $a<b$ then prove that

$$\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}$$ using

Lagrange's mean value theorem and deduce $\frac{\pi}{4} + \frac{3}{25} < \tan^{-1}\frac{4}{3} < \frac{\pi}{4} + \frac{1}{6}$

3. Verify Lagrange's mean value theorem for the function $f(x) = \cos x$ in $[0, \frac{\pi}{2}]$

4 Find 'c' of Cauchy's mean value theorem for $f(x) = \sqrt{x}$ and $g(x) = \frac{1}{\sqrt{x}}$ in $[a, b]$ where $0<a<b$

5 State the Taylor's series and expand the function $f(x) = \log(\sin x)$ in powers of $(x-2)$ by using Taylor's

6. State the maclaurin's series and obtain the maclaurin's series expansion of the function

$f(x) = \log(1+x)$

Unit 4

1. If $x^x y^y z^z = e$ then show that if $x = y = z$, $\frac{\partial^2 z}{\partial x \partial y} = -(x \log ex)^{-1}$

2. If $u = \frac{x+y}{1-xy}$, $v = \tan^{-1}x + \tan^{-1}y$ the find $\frac{\partial(u,v)}{\partial(x,y)}$. Hence prove that u and v are functionally dependent find the functional relation between them

3. A rectangular box open at the top is to have a volume of 32 cubic units. find the dimensions of the box requiring least material for its construction.

4. Divide 40 into three positive parts such that the product of the first part, cube of second part, and square of the third part is maximum

5. Express the integral

$$\int_0^{\frac{\pi}{2}} \sin^5\theta \cos^{\frac{7}{2}}\theta \, d\theta$$ in terms of

Beta function

6. Evaluate $\int_0^\infty x^4 e^{-x^2} \, dx$

7. Prove that

$$\int_0^1 x^{n-1} \left(\log \frac{1}{x}\right)^{m-1} dx = \frac{\Gamma(m)}{n^m}, \quad m, n > 0$$

Units

1. Evaluate $\int_0^5 \int_0^{x^2} x(x^2+y^2)\,dx\,dy$

2. Change of order of integration and evaluate $\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} dy\,dx$

3. Evaluate $\iint xy(x+y)\,dx\,dy$ over the region bounded by $y = x^2$ and $y = x$

4. Find the volume of the tetrahedron bounded by the planes $x=0$, $y=0$, $z=0$ and

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

5. Evaluate the triple integral $\iiint xyz\,dx\,dy\,dz$ over the positive octant of the sphere $x^2+y^2+z^2=a^2$

6 Evaluate the triple integral

$\iiint_V dx\,dy\,dz$ where V is the

finite region of space formed

by the planes $x=0, y=0, z=0$

and $2x+3y+4z=12$